Preliminary Engineering Calculation Report

Project	DEMO - Health post rooftop solar (sample data)
Client	Sample client (fictional)
Location	Kathmandu, Nepal
Prepared by	ElectricalCalc AI demo
Report date	2026-10-29
Software	ElectricalCalc AI v0.1.0
Review status	DRAFT - not reviewed
Reviewer	Not reviewed

This is a preliminary calculation report generated by software. It is not a certified, approved or stamped engineering design and must not be used for construction or procurement without review by a qualified engineer against site conditions, manufacturer data and the applicable local codes and regulations. Results are only as reliable as the inputs and assumptions listed.

Scope and notes

DEMONSTRATION ONLY. Client, consumption, tariff and costs are illustrative sample inputs, not a real project or real quotation. Solar data is a real PVGIS v5.3 response for Kathmandu (27° tilt, south).

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1. Solar PV preliminary sizing (hybrid with battery)

Input parameters

Location	Kathmandu, Nepal (demo)
Average daily consumption	18 kWh/day
Target share of consumption	100.00 %
Panel rating	550.00 Wp
Panel area	2.58 m²
DC/AC ratio	1.2
Available area	60 m²
Tariff	12 NPR/kWh
Export rate	0 NPR/kWh
Installed cost per kWp	95000 NPR/kWp
Backup energy required	6 kWh/day
Days of autonomy	1 days
Battery depth of discharge	90 %
Battery-to-AC efficiency	95 %
Battery bank voltage	48 V
Peak simultaneous load	3.5 kW
Battery cost	30000 NPR/kWh

Calculation method and equations

- E_demand = daily kWh × 365
- P_PV,required = E_demand × offset / Y (Y = specific yield, kWh/kWp/yr)
- N_panels = ceil(P_PV × 1000 / Wp_panel)
- E_gen = P_PV,installed × Y
- P_inverter,AC = P_PV / (DC:AC ratio)
- E_usable = backup kWh/day × autonomy days
- E_nominal = E_usable / (DoD × η_battery→AC)
- Ah = E_nominal × 1000 / V_bank
- Saving = min(E_gen, E_demand) × tariff + max(E_gen − E_demand, 0) × export rate
- Simple payback = capital cost / annual saving

Results (calculated by deterministic code)

Annual consumption (energy)	6,570.00 kWh/yr
PV capacity needed for target (power, DC)	4.605 kWp
Panels of 550 Wp	9 panels
Installed PV capacity (power, DC)	4.95 kWp
Module area	23.22 m²
Specific yield	1,426.64 kWh/kWp/yr
Estimated annual generation (energy, AC)	7,061.87 kWh/yr
Generation / consumption	107.49 %
Preliminary inverter AC rating (power)	4.125 kW
Required usable battery energy	6 kWh
Required nominal battery capacity	7.018 kWh
Nominal capacity at 48 V	146.20 Ah
Estimated annual bill saving	78,840.00 NPR/yr
Estimated installed cost (user-entered unit costs)	680,776.32 NPR

Simple payback	8.635 years

Monthly generation estimate

Month	kWh/kWp	Generation (kWh)	Consumption (kWh)
Jan	136.7	676.0	558.0
Feb	128.6	636.0	504.0
Mar	150.1	743.0	558.0
Apr	141.3	700.0	540.0
May	128.9	638.0	558.0
Jun	99.7	493.0	540.0
Jul	76.4	378.0	558.0
Aug	86.0	426.0	558.0
Sep	96.3	477.0	540.0
Oct	125.0	619.0	558.0
Nov	125.6	622.0	540.0
Dec	132.1	654.0	558.0

Preliminary bill of materials

Item	Specification	Quantity	Note
PV modules	550 Wp, ~2.58 m² each (user-entered)	9	Confirm model datasheet
Hybrid inverter	≈ 4.12 kW AC	1 (or split)	Check MPPT voltage window vs string design
Battery bank	≈ 7.0 kWh nominal @ 48 V	as per module size	Chemistry/DoD per datasheet
Mounting structure	To be sized in detailed design	-	
DC cables & connectors	To be sized in detailed design	-	
DC isolator / fuses	To be sized in detailed design	-	
AC breaker & isolator	To be sized in detailed design	-	
Surge protection (DC/AC)	To be sized in detailed design	-	
Earthing system	To be sized in detailed design	-	
Energy meter (net/bi-directional)	To be sized in detailed design	-	

Assumptions

- Solar resource: PVGIS v5.3 (PVGIS-ERA5, 2005-2023) at 27.7172, 85.3240; tilt 27°, azimuth 0° (0 = south); 14.0% system loss.
- Annual energy balance only - hourly mismatch between generation and consumption is not modelled.
- Inverter AC rating = PV DC capacity / DC-to-AC ratio of 1.2 (design choice; verify with the inverter manufacturer's string-sizing tool).
- Panel count rounded up; array layout, row spacing, shading and roof setbacks are not modelled.
- Available area is net usable module area (no allowance for spacing/access added).
- Usable battery energy limited by 90% depth of discharge (check battery warranty).
- 95% battery-to-AC conversion efficiency.
- Battery ageing, temperature derating and recharge time are not included.
- Flat tariff of 12 NPR/kWh with annual net metering; tariff slabs, demand charges, taxes, escalation and degradation are not included.

Warnings and limitations

- Motor/compressor start-up surge is not checked; confirm inverter surge rating.
- Preliminary estimate for feasibility only. Final design requires site survey, shading analysis, structural check and verification against local codes by a qualified engineer.

Data sources and references

- PVGIS (JRC, European Commission), https://re.jrc.ec.europa.eu/pvg_tools/

2. Inverter AC output current at 5 kW

Input parameters

Real power	5 kW
Voltage	230.00 V
Power factor	1

Calculation method and equations

- $I = P / (V \times \cos \varphi)$

Results (calculated by deterministic code)

Line current	21.74 A
Apparent power	5 kVA

Assumptions

- Balanced, sinusoidal load.

3. Inverter AC output cable - voltage drop check

Input parameters

System	single phase
Nominal voltage	230.00 V
Design current	21.7 A
One-way length	35 m
Conductor size	6 mm²
Conductor	copper
Conductor operating temperature	70 °C
Power factor	1
Reactance	0 mΩ/m

Calculation method and equations

- $\Delta V = 2 \times I \times L \times (R \cos \varphi + X \sin \varphi)$
- $R_T = R_{20} \times (1 + \alpha 20 \times (T - 20))$
- $\Delta V\% = \Delta V / V \times 100$

Results (calculated by deterministic code)

Conductor resistance at 70 °C	3.685 Ω/km
Voltage drop ΔV	5.598 V
Voltage drop	2.434 %
Voltage at load (approx.)	224.40 V
Conductor I²R loss	121.47 W
Within 3% limit	Yes

Assumptions

- Conductor resistance from IEC 60228 maximum values at 20 °C, corrected to 70 °C (α20 = 0.00393/K).
- Length is the one-way route length; the return conductor is included by the formula.
- Conductor reactance taken as zero (not entered).
- Sinusoidal current; skin and proximity effects neglected.

Warnings and limitations

- Voltage drop only: ampacity (installation method, ambient temperature, grouping, insulation), protective-device coordination and short-circuit withstand are not checked and must be verified against the applicable wiring rules.

Data sources and references

- IEC 60228:2004, Table 2 - maximum DC resistance of class 2 conductors at 20 °C

Review record

To be completed by the reviewing engineer. Signing below records review only; it does not by itself constitute certification under any jurisdiction's professional engineering rules.

Reviewed by	
Registration / licence no.	
Signature	
Date	
Comments	